The mugsthesis class

Paul D. Gessler
<paul@gessler.net>

v1.1, dated 2022/09/01

Abstract

mugsthesis is a document class to be used for typesetting Marquette University Graduate School (MUGS) theses and dissertations. It is based on the memoir class, and meets the basic requirements contained in the MUGS *Thesis/Dissertation Directives*.[1] The class provides a minimal set of changes to the memoir class to meet the university's specifications. As such, it should be compatible with any other packages that are compatible with the widely used memoir class.

1 Introduction

The mugsthesis document class is designed to allow straightforward preparation of MUGS theses and dissertations using the LaTeX document preparation system. It complies with the MUGS *Thesis/Dissertation Directives* while maintaining compatibility with other packages used to extend the functionality of LaTeX.

Development of the document class is currently hosted on sourcehut[2]. You are welcome to follow development, submit any issues you discover, or suggest improvements/enhancements using the web interface or mailing list there.

The easiest way to start using the mugsthesis document class is to take a look at the sample document provided, `mugsthesis_sample.tex`, along with its associated included files. You can take this as the basis for your own thesis or dissertation.

2 Package Options

The class supports several options for configuring the appearance of the thesis or dissertation. Apart from the options below, memoir is loaded with options `letterpaper` and `oneside` as specified in the *Thesis/Dissertation Directives*.

3 Document Type

The type of document, either a thesis or a dissertation, is adjusted through the use of the options `thesis` (default) and `dissertation`. `thesis` produces a thesis

[1] mugsthesis v1.1 is compatible with MUGS *Thesis Directives* dated 2014, available at https://www.marquette.edu/grad/documents/thesis-directives-2014.pdf and *Dissertation Directives* dated 2018, available at https://www.marquette.edu/grad/documents/dissertation-directives.pdf

[2] https://sr.ht/~pdgessler/mugsthesis/

and `dissertation` produces a dissertation. Currently, the only difference is in the text on the title page. The document type options are mutually exclusive.

3.1 Font Size

The base font size of the document is adjusted through use of the options `10pt`, `11pt`, and `12pt` (default). The font size options are mutually exclusive; if more than one is specified, the default will be used.

3.2 Other Options

The options `final` (default) and `draft` configure the output document: `final` produces a camera-ready copy of the thesis, while `draft` marks overfull lines with black rules in the margins, and may have interaction with other packages. The most common interaction is that graphicx will place empty boxes instead of the actual graphics when the `draft` option is set. These options are mutually exclusive.

The options `indentfirsttrue` (default) and `indentfirstfalse` configure the indentation of the first paragraph following a heading. The *Thesis Directives* show indentation of the initial paragraphs and specify that "new paragraphs should be indented 0.5 in". Centuries of typographers contend that the purpose of indentation is to mark the beginning of a new paragraph, and the heading itself serves that purpose for the first paragraph under a heading. So the `indentfirstfalse` option is provided if you agree with this contention. These options are mutually exclusive.

3.3 Unsupported memoir Options

The memoir class has many other options. Traditionally, an author can specify a base class option and the derived class will pass that option to the base class. However, many of these options would cause non-compliance with the MUGS *Thesis Directives*. For this reason, all unsupported options are reported to the log file and otherwise ignored.

4 Usage

A number of commands (macros) and environments are defined by mugsthesis for use in the preparation of theses. In-context examples of their use may be found in `mugsthesis_sample.tex`.

`\title` The `\title{`⟨*title*⟩`}` macro takes a single mandatory argument, which is the title of the thesis. Ensure that it is less than 120 characters long, and that it is entered with line breaks chosen to produce the requested inverted pyramid format. An example from the *Thesis Directives*:

```
\title{
  A STUDY OF THE SOCIOLOGICAL IMPACT OF THE\\
  1984 OLYMPICS ON THE POVERTY\\
  LEVEL OF CITIZENS OF\\
  LOS ANGELES
}
```

The capitalization here is not strictly necessary; all letters are converted to upper case internally.

`\author` The macro `\author{`⟨*author*⟩`}` accepts a single mandatory argument, which should contain the author's name and credentials. Using the *Thesis Directives* example once again:

```
\author{John J. Smith, B.A.}
```

`\degree` The `\degree{`⟨*degree*⟩`}` macro takes a single mandatory argument, the degree pursued by the author. An example from the directives, as ever:

```
\degree{Master of Science}
```

`\degreemo` `\degreeyr` The macros `\degreemo{`⟨*month*⟩`}` and `\degreeyr{`⟨*year*⟩`}` each accept a single argument, which is the month and year, respectively, during which the degree is to be conferred. The value of ⟨*month*⟩ must be one of May, August, or December:

```
\degreemo{December}
\degreeyr{2013}
```

`\maketitle` The `\maketitle` command typesets the title page with all the previously-declared metadata, and should be used immediately at the beginning of the `document` environment.

`abstract` (*env.*) `acknowledgments` (*env.*) `dedication` (*env.*) The environments `abstract` (mandatory), `acknowledgments` (optional), and `dedication` (optional) should be used for the respective content, if included. The `abstract` environment must immediately follow `\maketitle`, and the `acknowledgments` and `dedication` environments, if used, should follow `abstract`.

`\tableofcontents` `\listoftables` `\listoffigures` The Graduate School requires a table of contents to appear in every thesis. Additionally, lists of tables and lists of figures may be used. The commands `\tableofcontents`, `\listoftables`, and `\listoffigures` should be used to generate these lists with the appropriate formatting. `\tableofcontents` should follow the abstract and any optional environments used in the front matter. `\listoftables` or `\listoffigures` should be used after `\tableofcontents`. If both are used, `\listoftables` should precede `\listoffigures`.

`\frontmatter` `\mainmatter` `\appendix` The switch commands `\frontmatter`, `\mainmatter`, and `\appendix` set the page and chapter numbering styles for the appropriate parts of the thesis. They should immediately precede the abstract, first main body chapter, and first appendix (if any), respectively.

`\bibname` The `\bibname` command contains the name text for the bibliography (default: BIBLIOGRAPHY). If the chosen citation/reference style specifies a different name (for example: REFERENCES), it can be changed with:

```
\renewcommand{\bibname}{REFERENCES}
```

Many other commands and environments are defined by the memoir class for document elements including (sub)figures, (sub)tables, associated captions, quotations, and more. Consult the memoir documentation[3] for complete details. If these or customizations from other packages are used, it is ultimately the author's responsibility to verify that compliance with the *Thesis Directives* is maintained.

5 Implementation

First, load etoolbox for patching existing commands.

```
1 \RequirePackage{etoolbox}
```

[3] http://texdoc.net/pkg/memoir

Declare a new boolean to control indentation of initial paragraphs.

```
\newbool{indentfirst}
```

Declare class options, pass them to the base class if applicable, and discard any unsupported options, warning the user they have been ignored.

```
\DeclareOption{thesis}{\def\@mugsdoctype{Thesis}}
\DeclareOption{dissertation}{\def\@mugsdoctype{Dissertation}}
\DeclareOption{10pt}{\PassOptionsToClass{10pt}{memoir}}
\DeclareOption{11pt}{\PassOptionsToClass{11pt}{memoir}}
\DeclareOption{12pt}{\PassOptionsToClass{12pt}{memoir}}
\DeclareOption{final}{\PassOptionsToClass{final}{memoir}}
\DeclareOption{draft}{\PassOptionsToClass{draft}{memoir}}
\DeclareOption{indentfirsttrue}{\booltrue{indentfirst}}
\DeclareOption{indentfirstfalse}{\boolfalse{indentfirst}}
\DeclareOption*{%
  \ClassWarning{mugsthesis}{Unknown option `\CurrentOption' ignored}%
}
```

Execute default options `thesis`, `12pt`, `final`, and `indentfirsttrue`, and process the author-specified options.

```
\ExecuteOptions{thesis,12pt,final,indentfirsttrue}
\ProcessOptions\relax
```

Use the indentfirst package if `indentfirsttrue` is set.

```
\ifbool{indentfirst}{\RequirePackage{indentfirst}}{}
```

Load the base memoir class with options `letterpaper` and `oneside`.

```
\LoadClass[letterpaper,oneside]{memoir}
```

Set the line spacing and justification according to the MUGS *Thesis Directives*. Additionally set `\raggedbottom` to avoid underfull `\hbox`es with rigid skip lengths.

```
\DoubleSpacing
\raggedright
\raggedbottom
```

Set the page layout.

```
\setlrmarginsandblock{1.5in}{1.0in}{*} % left and right margins
\setulmarginsandblock{1.0in}{1.0in}{*} % top and bottom margins
\setheaderspaces{0.5in}{*}{*} % header 0.5in from edge
\marginparmargin{left} % use the wider (left) margin for notes
\setmarginnotes{0.125in}{1.25in}{\baselineskip} % type area for notes
\checkandfixthelayout[lines] % check the layout; TODO check algorithm selection
```

Set the paragraph indentation.

```
\setlength\parindent{0.5in}
```

Define and use the `mugs` pagestyle for page numbering throughout the thesis.

```
\makepagestyle{mugs}
\makeevenhead{mugs}{}{}{\thepage}
\makeoddhead{mugs}{}{}{\thepage}
```

Alias the `chapter` pagestyle to `mugs` so the chapter opening pages have numbering identical to the rest of the pages, and set the page style for the document.

```
\aliaspagestyle{chapter}{mugs}
\pagestyle{mugs}
```

`\degree` `\degreemo` `\degreeyr`

Define `\degree{`⟨*degree*⟩`}` to set the degree name for the title page.

```
\newcommand{\degree}[1]{\def\mugsdegree{#1}}
```

Similarly, define `\degreemo{`⟨*month*⟩`}` and `\degreeyr{`⟨*year*⟩`}` to set the date of conferral for the title and abstract pages.

```
\newcommand{\degreemo}[1]{\def\mugsdegreemo{#1}}
\newcommand{\degreeyr}[1]{\def\mugsdegreeyr{#1}}
```

`\title` Redefine `\title` to use full upper case.

```
\renewcommand{\title}[1]{\def\@title{\uppercase{#1}}}
```

`\maketitle` Redefine `\maketitle` to produce the appropriate title page, using all author metadata.

```
\renewcommand{\maketitle}{{%
  \cleardoublepage
  \pagenumbering{Alph}
  \SingleSpace
  \thispagestyle{empty}
  \centering
  \@title
  \vfill
  by
  \vskip \onelineskip
  \@author
  \vfill
  A \@mugsdoctype\ submitted to the Faculty of the Graduate School,\\
  Marquette University,\\
  in Partial Fulfillment of the Requirements for\\
  the Degree of \mugsdegree{}
  \vfill
  Milwaukee, Wisconsin
  \vskip \onelineskip
  \mugsdegreemo~\mugsdegreeyr
  \clearpage
}}
```

`abstract` (*env.*) Redefine the `abstract` environment for typesetting according to the directives.

```
\renewenvironment{abstract}{%
  \thispagestyle{empty}
  \begin{SingleSpace}
  \begin{center}
    ABSTRACT\\
    \@title
    \vskip 2\onelineskip
    \@author
    \vskip \onelineskip
    Marquette University, \mugsdegreeyr
    \vskip 2\baselineskip
  \end{center}
  \@afterindentfalse\@afterheading % makes indentfirst work with custom envs
}{
  \end{SingleSpace}\setcounter{page}{0}
}
```

`acknowledgments` (*env.*) Define the optional `acknowledgments` environment.

```
\newenvironment{acknowledgments}{%
```

```
  \addcontentsline{toc}{chapter}{ACKNOWLEDGMENTS}
  \chapter*{Acknowledgments}
  \begin{SingleSpace}
  \begin{center}
    \unskip
    \@author
    \vskip 2\onelineskip
  \end{center}
  \@afterindentfalse\@afterheading
}{\end{SingleSpace}}
```

`dedication` (*env.*) Define the optional `dedication` environment.

```
\newenvironment{dedication}{%
  \addcontentsline{toc}{chapter}{DEDICATION}
  \chapter*{Dedication}
  \begin{SingleSpace}\vskip\onelineskip
  \@afterindentfalse\@afterheading
}{\end{SingleSpace}}
```

Declare subfloats using memoir facilities.[4] Ensure any subfloats are listed in the list of figures and list of tables.

```
\newsubfloat{figure}
\newsubfloat{table}
\setcounter{lofdepth}{2}
\setcounter{lotdepth}{2}
```

Define and use the MUGS chapter style throughout the thesis.

```
\makechapterstyle{mugs}{%
  \setlength{\midchapskip}{0pt}
  \setlength{\afterchapskip}{0pt}
  \renewcommand*{\chapterheadstart}{}
  \renewcommand*{\chaptitlefont}{\bfseries}
  \renewcommand*{\printchaptername}{\centering \MakeTextUppercase{\@chapapp}}
  \renewcommand*{\printchapternum}{\thechapter}
  \renewcommand*{\printchaptertitle}[1]{%
    \SingleSpacing \protect\parbox{\textwidth}{%
      \centering \chaptitlefont \MakeTextUppercase{##1}
    }
  }
}% end{mugs}
\chapterstyle{mugs}
```

Put the LoT and LoF on their own pages by patching their definitions.

```
\pretocmd{\listoftables}{\clearforchapter}{}{%
  \ClassWarning{mugsthesis}{Patching `\noexpand\listoftables' failed}
}
```

```
\pretocmd{\listoffigures}{\clearforchapter}{}{%
  \ClassWarning{mugsthesis}{Patching `\noexpand\listoffigures' failed}
}
```

`\contentsname` `\listtablename` `\listfigurename` `\bibname` Style the ToC, LoT, LoF, and Bibliography headings.

```
\renewcommand*{\contentsname}{TABLE OF CONTENTS}
```

[4] These are not needed by all users; consider an option to enable? Maximum depth required?

```
\renewcommand*{\listtablename}{LIST OF TABLES}
\renewcommand*{\listfigurename}{LIST OF FIGURES}
\renewcommand*{\bibname}{BIBLIOGRAPHY}
```

Specify one line skip between bibliography entries.

```
\setlength{\bibitemsep}{\onelineskip}
```

`\prebibhook` Redefine `\prebibhook` to set single spacing in the bibliography and proper spacing after the heading.

```
\renewcommand*{\prebibhook}{\SingleSpacing\vskip\onelineskip}
```

`\biblistextra` Redefine `\biblistextra` to set item separation and indentation.

```
\renewcommand*{\biblistextra}{%
  \setlength{\itemsep}{\bibitemsep}
  \setlength{\leftmargin}{0.375in}% hanging indent
  \setlength{\itemindent}{\labelwidth}%
  \addtolength{\itemindent}{\labelsep}
  \addtolength{\itemindent}{-\leftmargin}
}
```

`\tableofcontents` Do not place an entry for the ToC within the ToC itself.

```
\renewcommand{\tableofcontents}{%
  \@ifstar{\mem@tableofcontents{01}}
          {\mem@tableofcontents{01}}%
}
```

Set the indentation for the ToC, LoT, and LoF (0.5 in additional per level).

```
\setlength{      \cftchapterindent}{0.0in}
\setlength{      \cftsectionindent}{0.5in}
\setlength{   \cftsubsectionindent}{1.0in}
\setlength{\cftsubsubsectionindent}{1.5in}
\setlength{    \cftparagraphindent}{2.0in}
\setlength{ \cftsubparagraphindent}{2.5in}
\setlength{        \cfttableindent}{0.0in}
\setlength{     \cftsubtableindent}{0.5in}
\setlength{       \cftfigureindent}{0.0in}
\setlength{    \cftsubfigureindent}{0.5in}
```

Ragged right within and one line skip between ToC, LoT, and LoF entries.

```
\setrmarg{3.55em plus 1fil}
\setlength{\cftbeforechapterskip}{\onelineskip}
\setlength{\cftbeforesectionskip}{\onelineskip}
\setlength{\cftbeforesubsectionskip}{\onelineskip}
\setlength{\cftbeforesubsubsectionskip}{\onelineskip}
\setlength{\cftbeforeparagraphskip}{\onelineskip}
\setlength{\cftbeforesubparagraphskip}{\onelineskip}
\setlength{\cftbeforetableskip}{\onelineskip}
\setlength{\cftbeforesubtableskip}{\onelineskip}
\setlength{\cftbeforefigureskip}{\onelineskip}
\setlength{\cftbeforesubfigureskip}{\onelineskip}
```

`\insertchapterspace` No extra space between chapters.

```
\renewcommand{\insertchapterspace}{}
```

\aftertoctitle
\afterlottitle
\afterloftitle

Switch to single spacing in the ToC, LoT, and LoF.

```
\renewcommand{\aftertoctitle}{\vspace{-\onelineskip}\aftergroup\SingleSpacing}
\renewcommand{\afterlottitle}{\vspace{-\onelineskip}\aftergroup\SingleSpacing}
\renewcommand{\afterloftitle}{\vspace{-\onelineskip}\aftergroup\SingleSpacing}
```

\cftchapterfont
\cftchapterleader
\cftchapterpagefont

Style the ToC chapter headings.

```
\renewcommand{\cftchapterfont}{\normalfont}
\renewcommand{\cftchapterleader}{\cftsectionleader}
\renewcommand{\cftchapterpagefont}{\normalfont}
```

\f@rtoc

For hyperref bookmarking functionality (if present).

```
\AtEndPreamble{\providecommand{\texorpdfstring}[2]{#1}}
\settocpreprocessor{chapter}{%
\let\tempf@rtoc\f@rtoc%
\def\f@rtoc{%
  \texorpdfstring{\MakeTextUppercase{\tempf@rtoc}}{\tempf@rtoc}}%
}
```

Set spacing in quote and quotation environments.

```
\AtBeginEnvironment{quote}{\par\SingleSpacing}
\AtBeginEnvironment{quotation}{\par\SingleSpacing}
```

Set sectional unit numbering levels and ToC depth.

```
\maxsecnumdepth{subsection}
\settocdepth{subsection}
```

Style the lower-level headings.

```
\newlength{\preheadskip}\setlength{\preheadskip}{1sp plus 1sp minus 1sp}
\newlength{\postheadskip}\setlength{\postheadskip}{\onelineskip}
\setsecheadstyle{\bfseries\SingleSpacing}
\setbeforesecskip{\preheadskip}
\setaftersecskip{\postheadskip}
\setsubsecheadstyle{\bfseries\SingleSpacing}
\setbeforesubsecskip{\preheadskip}
\setaftersubsecskip{\postheadskip}
\setsubsubsecheadstyle{\bfseries\SingleSpacing}
\setbeforesubsubsecskip{\preheadskip}
\setaftersubsubsecskip{\postheadskip}
```

Set the list and table spacings.

```
\tightlists
\AtBeginEnvironment{tabular}{\SingleSpacing}
```

Set the caption styles and spacing.

```
\captionstyle[\centering]{\raggedright}
\subcaptionsize{\small}
\setlength{\abovecaptionskip}{\parsep}
\setlength{\belowcaptionskip}{-\parsep}
```

Avoid orphans, widows, and other detritus.

```
\clubpenalty=10000
\widowpenalty=9996
\brokenpenalty=4991
\predisplaypenalty=10000
\postdisplaypenalty=1549
\displaywidowpenalty=1602
```

Set the spacing between floats and surrounding elements.

```
195 \setlength{\floatsep}{3\onelineskip}
196 \setlength{\intextsep}{2\onelineskip}
197 \setlength{\textfloatsep}{2\onelineskip}
198 \setlength{\dblfloatsep}{2\onelineskip}
199 \setlength{\dbltextfloatsep}{2\onelineskip}
```

Change History

v1.0

General: Initial public release 1

v1.1

General: Added document type options for theses and dissertations 1

Index

Numbers written in italic refer to the page where the corresponding entry is described; numbers underlined refer to the code line of the definition; numbers in roman refer to the code lines where the entry is used.

Symbols

\@afterheading 72, 85, 91
\@afterindentfalse 72, 85, 91
\@author 48, 67, 82
\@chapapp 102
\@mugsdoctype . . 3, 4, 50
\␣ 50

A

\abovecaptionskip . . 187
abstract (env.) . . . *3*, 60
acknowledgments (env.) *3*, 76
\afterchapskip 99
\afterloftitle 156
\afterlottitle 156
\aftertoctitle 156
\aliaspagestyle . . . 32
\appendix *3*
\author *3*

B

\belowcaptionskip . . 188
\bibitemsep . . . 121, 124
\biblistextra 123
\bibname *3*, 117
\brokenpenalty 191

C

\captionstyle 185
\cftbeforechapterskip 145
\cftbeforefigureskip 153
\cftbeforeparagraphskip 149
\cftbeforesectionskip 146
\cftbeforesubfigureskip 154
\cftbeforesubparagraphskip 150
\cftbeforesubsectionskip 147
\cftbeforesubsubsectionskip 148
\cftbeforesubtableskip 152
\cftbeforetableskip 151
\cftchapterfont . . . 159
\cftchapterindent . . 134
\cftchapterleader . . 159
\cftchapterpagefont 159
\cftfigureindent . . . 142
\cftparagraphindent 138
\cftsectionindent . . 135
\cftsectionleader . . 160
\cftsubfigureindent 143
\cftsubparagraphindent 139
\cftsubsectionindent 136
\cftsubsubsectionindent 137
\cftsubtableindent . 141
\cfttableindent . . . 140
\chapter 78, 89
\chapterheadstart . . 100
\chapterstyle 110
\chaptitlefont 101, 106
\checkandfixthelayout 27
\ClassWarning 13, 112, 115
\cleardoublepage . . . 39
\clearforchapter 111, 114
\clearpage 58
\clubpenalty 189
\contentsname 117
\CurrentOption 13

D

\dblfloatsep 198
\dbltextfloatsep . . . 199
\DeclareOption 3, 4, 5, 6, 7, 8, 9, 10, 11, 12
dedication (env.) . . *3*, 87
\degree *3*, 34
\degreemo *3*, 34

\degreeyr *3*, 34
\displaywidowpenalty 194
\DoubleSpacing 19

E

environments:
abstract *3*, 60
acknowledgments *3*, 76
dedication *3*, 87
\ExecuteOptions . . . 15

F

\f@rtoc 162
\floatsep 195
\frontmatter *3*

I

\insertchapterspace 155
\intextsep 196
\itemindent 126, 127, 128
\itemsep 124

L

\labelsep 127
\labelwidth 126
\leftmargin . . . 125, 128
\listfigurename . . . 117
\listoffigures *3*, 114, 115
\listoftables *3*, 111, 112
\listtablename 117
\LoadClass 18

M

\mainmatter *3*
\makechapterstyle . . 97
\makeevenhead 30
\makeoddhead 31
\makepagestyle 29
\MakeTextUppercase 102, 106, 166
\maketitle *3*, 38
\marginparmargin . . . 25
\maxsecnumdepth . . . 170
\mem@tableofcontents 131, 132
\midchapskip 98
\mugsdegree 34, 53
\mugsdegreemo . . . 35, 57
\mugsdegreeyr 36, 57, 69

N

\newsubfloat 93, 94

P

\pagenumbering 40
\pagestyle 33
\parindent 28
\parsep 187, 188
\PassOptionsToClass 5, 6, 7, 8, 9
\postdisplaypenalty 193
\postheadskip 173, 176, 179, 182
\prebibhook 122
\predisplaypenalty . 192
\preheadskip 172, 175, 178, 181
\printchaptername . . 102
\printchapternum . . . 103
\printchaptertitle . 104
\ProcessOptions . . . 16

R

\raggedbottom 21
\raggedright . . . 20, 185
\relax 16

S

\setaftersecskip . . . 176
\setaftersubsecskip 179
\setaftersubsubsecskip 182
\setbeforesecskip . . 175
\setbeforesubsecskip 178
\setbeforesubsubsecskip 181
\setheaderspaces . . . 24
\setlrmarginsandblock 22
\setmarginnotes . . . 26
\setrmarg 144
\setsecheadstyle . . . 174
\setsubsecheadstyle 177
\setsubsubsecheadstyle 180
\settocdepth 171
\settocpreprocessor 163
\setulmarginsandblock 23
\SingleSpace 41
\SingleSpacing . 105, 122, 156, 157, 158, 168, 169, 174, 177, 180, 184
\subcaptionsize . . . 186

T

\tableofcontents . *3*, 130
\tempf@rtoc . . . 164, 166
\textfloatsep 197
\textwidth 105
\thechapter 103
\thepage 30, 31
\thispagestyle . . 42, 61
\tightlists 183
\title *2*, 37

U

\uppercase 37

W

\widowpenalty 190